

Sample Outputs for basic Multit-armed bandit simulation

Algorithm : Incremental Action value approach

$$(Q_{\text{new}} = Q_{\text{old}} + \alpha * (R - Q_{\text{old}}))$$

Where $\alpha = 1/N$

Intuition: agent says, " use the most correctly seemed option, yet discover new options sometimes "

Sample Run 1: At epsilon value = 0.2

```
Exploiting...
... Step: 0
Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.0, 0, 0, 0]
-----
Exploiting...
Step: 1
Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.0, 0, 0, 0]
-----
Exploiting...
Step: 2
Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.0, 0, 0, 0]
-----
Exploiting...
Step: 3
Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.0, 0, 0, 0]
-----
Exploiting...
Step: 4
Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.0, 0, 0, 0]
-----
Exploiting...
Step: 5
```

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*** -----
Exploring...
Step: 40
Chosen Bandit: 1
Reward: 1
Estimated Rewards: [0.125, 0.6, 0.8571428571428571, 0]
-----
Exploiting...
Step: 41
Chosen Bandit: 2
Reward: 1
Estimated Rewards: [0.125, 0.6, 0.8620689655172413, 0]
-----
Exploring...
Step: 42
Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.1111111111111111, 0.6, 0.8620689655172413, 0]
-----
Exploiting...
Step: 43
Chosen Bandit: 2
Reward: 1
Estimated Rewards: [0.1111111111111111, 0.6, 0.8666666666666666, 0]
-----
Exploiting...
Step: 44
Chosen Bandit: 2
Reward: 1
Estimated Rewards: [0.1111111111111111, 0.6, 0.8709677419354838, 0]

```

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Step: 95
Chosen Bandit: 2
*** Reward: 1
Estimated Rewards: [0.0909090909090909, 0.6666666666666666, 0.8243243243243243, 0.6]
-----
Exploring...
Step: 96
Chosen Bandit: 0
Reward: 1
Estimated Rewards: [0.1666666666666666, 0.6666666666666666, 0.8243243243243243, 0.6]
-----
Exploiting...
Step: 97
Chosen Bandit: 2
Reward: 1
Estimated Rewards: [0.1666666666666666, 0.6666666666666666, 0.8266666666666667, 0.6]
-----
Exploring...
Step: 98
Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.15384615384615383, 0.6666666666666666, 0.8266666666666667, 0.6]
-----
Exploiting...
Step: 99
Chosen Bandit: 2
Reward: 0
Estimated Rewards: [0.15384615384615383, 0.6666666666666666, 0.8157894736842105, 0.6]
-----
Total Reward: 71

```

Sample Run 2: At epsilon value = 0.2

```
Exploring...
Step: 0
*** Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.0, 0, 0, 0]
-----
Exploiting...
Step: 1
Chosen Bandit: 0
Reward: 1
Estimated Rewards: [0.5, 0, 0, 0]
-----
Exploiting...
Step: 2
Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.3333333333333337, 0, 0, 0]
-----
Exploiting...
Step: 3
Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.25, 0, 0, 0]
-----
Exploiting...
Step: 4
Chosen Bandit: 0
Reward: 0
Estimated Rewards: [0.2, 0, 0, 0]
-----
Exploring...
Step: 5
***
```

```
*** -----
Exploiting...
Step: 96
Chosen Bandit: 2
Reward: 0
Estimated Rewards: [0.14285714285714285, 0.4, 0.743243243243243, 0.25]
-----
Exploiting...
Step: 97
Chosen Bandit: 2
Reward: 0
Estimated Rewards: [0.14285714285714285, 0.4, 0.7333333333333331, 0.25]
-----
Exploiting...
Step: 98
Chosen Bandit: 2
Reward: 0
Estimated Rewards: [0.14285714285714285, 0.4, 0.7236842105263155, 0.25]
-----
Exploiting...
Step: 99
Chosen Bandit: 2
Reward: 1
Estimated Rewards: [0.14285714285714285, 0.4, 0.727272727272727, 0.25]
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Total Reward: 61
```